

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The correct answer is true for Statement 1 because ImageNet indeed contains images of various resolutions, while Statement 2 is false because ImageNet has significantly more images than Caltech-101, making it the larger dataset.

2. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Both statements are correct because Word 2 Vec utilizes a neural network architecture without the initialization from a Restricted Boltzmann Machine, and the tanh function is indeed a nonlinear activation function commonly used in neural networks.

3. **Answer:** C

Options: A) 0; B) 1; C) 2; D) 3

Note: The dimensionality of the null space of the matrix A is 2 because its rank is 1, resulting in a nullity of 2 as determined by the Rank-Nullity Theorem, which states that the dimension of the null space plus the rank must equal the number of columns.

4. **Answer:** D

Options: A) L_0 norm; B) L_1 norm; C) L_2 norm; D) either (a) or (b)

Note: The correct answer is either (a) or (b) because both Lasso (L_1) regularization and Ridge (L_2) regularization can lead to zero coefficients, but it is specifically Lasso that is known for producing sparse solutions by forcing some coefficients to be exactly zero.

5. **Answer:** D

Options: A) Increase bias; B) Decrease bias; C) Increase variance; D) Decrease variance

Note: Averaging the output of multiple decision trees reduces variance by mitigating the impact of individual tree overfitting, leading to a more stable and accurate overall prediction.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Residual connections help mitigate the vanishing gradient problem by allowing gradients to flow through the network more effectively during backpropagation, enabling the successful training of deeper architectures like ResNets and Transformers.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the correct expression for the joint probability $P(A, B, C)$ is $P(A | B, C) * P(B | C) * P(C)$, not the product of the conditional probabilities given in the statement, which do not account for the correct conditioning of variables.

8. **Answer:** True

Options: A) True; B) False

Note: Underfitting occurs when a model is too simplistic to represent the underlying relationships in the training data, leading to inadequate learning and consequently poor performance on both the training set and unseen data.

9. **Answer:** False

Options: A) True; B) False

Note: The trade-off between underfitting and overfitting is primarily influenced by model complexity and the amount of training data, rather than the method used to learn the weights.

10. **Answer:** True

Options: A) True; B) False

Note: Clustering effectively identifies fraudulent credit card transactions by grouping similar transaction patterns, allowing for the detection of anomalies that deviate from established norms, which are often indicative of fraud.

Long Form Response

11. **Answer:** The statements regarding entropy in joint distributions and the effects of moralization on directed graphs are both false. Entropy, as a measure of uncertainty, does not inherently apply to joint distributions in the manner implied; rather, it is context-dependent and influenced by the relationships between variables. Additionally, moralization, which involves creating an undirected graph from a directed graph by adding edges between parents of common children, does not always lead to accurate representations of dependencies in probabilistic models. In machine learning, these misconceptions can lead to flawed model assumptions and suboptimal learning outcomes, emphasizing the need for careful consideration of graphical representations and their implications for entropy. Thus, understanding these concepts correctly is crucial for effective application in machine learning contexts.
- Note:* correct because entropy is context-dependent and does not universally apply to joint distributions, while moralization can distort relationships in directed graphs, leading to potential inaccuracies in representing dependencies, which is critical in machine learning applications.
12. **Answer:** The Naive Bayes algorithm operates under the assumption of attribute independence, meaning that it presumes that the presence or absence of a particular

feature (or attribute) in the data is independent of the presence or absence of any other feature, given the class value. This assumption simplifies the computation of the posterior probabilities, allowing for efficient classification. However, in real-world scenarios, features are often correlated, which can lead to a violation of this independence assumption. As a result, the model may perform suboptimally, as it may underestimate or overestimate the importance of certain attributes that interact with one another. Consequently, while Naive Bayes can be effective in many applications, its reliance on this strong independence assumption can limit its accuracy in datasets where attributes are statistically dependent.

Note: correct because it accurately describes the fundamental assumption of attribute independence in the Naive Bayes algorithm, explaining how this simplification facilitates efficient probability calculations while also acknowledging the potential impact of feature correlation on model performance in practical applications.

End of Answer Key